

Apparent Dynamics of a Free Geodesic Projected from a Sphere onto a Plane

A worked derivation in classical mechanics and differential geometry

Andrea Sanna

Abstract

A point B moves freely (along a geodesic) on a sphere $S^2(R) \subset \mathbb{R}^3$. An observer A confined to the equatorial plane $z = 0$ sees only the orthogonal projection $B' = (x, y)$ of B . We ask what apparent dynamics A reconstructs. The construction hinges on the *time map* $dt/d\lambda$ relating the observer's time t to the affine parameter λ along the geodesic. Of three candidate maps, exactly one is physically admissible; it coincides with the special-relativistic time-dilation rule. For that map we derive in closed form the apparent velocity and acceleration, prove that the observed speed is conserved, and exhibit an exact invariant $|a| f^{3/2} = R\omega^2 \cos i$. Every analytical result is cross-verified numerically to machine precision. This note documents the derivation; the accompanying Python scripts reproduce all figures and checks.

1 Setup

Definition 1 (Spaces and parameters). *Let $\mathbb{R}^3 = (x, y, z)$ with the Euclidean metric, and let the observer A live in the plane $z = 0$. The point B moves on the sphere $S^2(R)$ of radius R . We write λ for the affine parameter along the geodesic, t for the observer's coordinate time, τ for B 's proper time, i for the inclination of the great circle to the plane, $\phi = \omega\lambda$ for the angular position, $\omega = V/R$ for the (constant) angular velocity, and $V = R\omega$ for B 's speed.*

2 The geodesic and its projection

Proposition 1 (Great circle). *A geodesic on $S^2(R)$ of inclination i , with ascending node on the x -axis, is*

$$x = R \cos \phi, \quad y = R \sin \phi \cos i, \quad z = R \sin \phi \sin i. \quad (1)$$

Proof. $x^2 + y^2 + z^2 = R^2[\cos^2 \phi + \sin^2 \phi(\cos^2 i + \sin^2 i)] = R^2$, and the curve is the intersection of $S^2(R)$ with a plane through the origin, hence a great circle. $\square$

Differentiating gives the velocity in $\mathbb{R}^3$,

$$\dot{x} = -V \sin \phi, \quad \dot{y} = V \cos \phi \cos i, \quad \dot{z} = V \cos \phi \sin i, \quad (2)$$

with the invariant $\dot{x}^2 + \dot{y}^2 + \dot{z}^2 = V^2$.

Definition 2 (Projection). *The observer sees $\mathbf{r}_{B'} = (x, y) = (R \cos \phi, R \sin \phi \cos i)$, an ellipse with semi-axes $a = R$ and $b = R \cos i$.*

3 The fundamental factor, and why $f \neq g$

Definition 3 (Velocity and position factors).

$$f(\phi) \equiv 1 - \frac{\dot{z}^2}{V^2} = 1 - \cos^2 \phi \sin^2 i = \frac{|\mathbf{V}_{R2}|^2}{V^2}, \quad (3)$$

$$g(\phi) \equiv \frac{|\mathbf{r}_{B'}|^2}{R^2} = 1 - \sin^2 \phi \sin^2 i. \quad (4)$$

These are *distinct* functions, related by a quarter-period phase shift,

$$f(\phi) = g\left(\phi + \frac{\pi}{2}\right), \quad f + g = 2 - \sin^2 i = \text{const.} \quad (5)$$

The factor f governs the velocity (how much speed remains visible after the hidden z -motion is removed); g governs position. Conflating them is the most common error in this projection, and it is the reason the apparent force does not reduce to a simple power law in $r = |\mathbf{r}_{B'}|$.

4 The time map: three candidates, one admissible

The observer's clock is fixed by the choice of $dt/d\lambda$. We examine three.

- **Option A:** $dt/d\lambda = \dot{z}/V = \cos \phi \sin i$. This changes sign at $\phi = \pi/2$: the clock would run backwards over half the orbit. Projected velocities also diverge. *Not physical.*
- **Option C:** $dt/d\lambda = |\dot{z}|/V$. The absolute value removes the sign change but creates zeros at the poles, where the clock stops and velocities diverge. *Not physical.*
- **Option B:** $dt/d\lambda = \sqrt{1 - \dot{z}^2/V^2} = \sqrt{f}$. Positive everywhere, smooth, and equal to the special-relativistic factor $\sqrt{1 - \beta_z^2}$ with $\beta_z = \dot{z}/V$. *The unique admissible choice.*

5 Closed-form dynamics under Option B

Proposition 2 (Generalised Lorentz factor and velocity). *With $\gamma \equiv d\lambda/dt = 1/\sqrt{f}$, the apparent velocity is*

$$u_x = \frac{\dot{x}}{\sqrt{f}} = -\frac{V \sin \phi}{\sqrt{f}}, \quad u_y = \frac{\dot{y}}{\sqrt{f}} = \frac{V \cos \phi \cos i}{\sqrt{f}}. \quad (6)$$

Theorem 1 (Conservation of observed speed). $|\mathbf{V}_{R2}| = \sqrt{u_x^2 + u_y^2} = V$, *constant around the orbit.*

Proof. $u_x^2 + u_y^2 = V^2(\sin^2 \phi + \cos^2 \phi \cos^2 i)/f = V^2 f/f = V^2$. □

Applying the transformation rule $\frac{d^2 x_i}{dt^2} = \frac{\ddot{x}_i}{f} - \frac{\dot{x}_i \dot{f}}{2f^2}$ (obtained by differentiating $u_i = \dot{x}_i/\sqrt{f}$ once more in t) together with the geodesic relation $\ddot{x}_i = -\omega^2 x_i$ and $\dot{f} = 2\omega^2 z \dot{z}/V^2$, the cross terms cancel exactly:

$$\frac{d^2 x}{dt^2} = -\frac{\omega^2 \cos^2 i}{f^2} x, \quad (7)$$

$$\frac{d^2 y}{dt^2} = -\frac{\omega^2}{f^2} y. \quad (8)$$

In matrix form,

$$\frac{d^2 \mathbf{r}_{B'}}{dt^2} = -\frac{\omega^2}{f^2} D \mathbf{r}_{B'}, \quad D = \begin{pmatrix} \cos^2 i & 0 \\ 0 & 1 \end{pmatrix}. \quad (9)$$

The apparent “force” is a restoring, anisotropic field modulated by $1/f^2$; it is not central for $i \neq 0$, and because $f \neq r^2/R^2$ it is not a power law in r .

6 An exact invariant

Theorem 2 (Invariant of the projected motion).

$$|a| f^{3/2} = R\omega^2 \cos i = \text{const.} \quad (10)$$

Proof. From the closed forms, $|a|^2 = \frac{R^2 \omega^4 \cos^2 i}{f^4} (\cos^2 \phi \cos^2 i + \sin^2 \phi) = \frac{R^2 \omega^4 \cos^2 i}{f^4} f = \frac{R^2 \omega^4 \cos^2 i}{f^3}$, using $\cos^2 \phi \cos^2 i + \sin^2 \phi = 1 - \cos^2 \phi \sin^2 i = f$. Taking the square root and multiplying by $f^{3/2}$ gives the result. $\square$

This is verified numerically to a standard deviation of order 10^{-15} across all inclinations $i \in [0^\circ, 89^\circ]$.

7 Time dilation

With units in which V plays the role of a limiting speed,

$$\frac{dt}{d\tau} = \sqrt{f} = \sqrt{1 - \beta_z^2}, \quad \beta_z = \frac{\dot{z}}{V}, \quad (11)$$

which is exactly the special-relativistic time-dilation relation. Thus a purely geometric projection of free motion reproduces, for the admissible time map, both an effective Lorentz factor $\gamma = 1/\sqrt{f}$ and SR-type time dilation.

8 Numerical verification

The accompanying scripts implement the construction and confirm, to machine precision: (i) the closed-form acceleration matches a central-difference numerical derivative; (ii) the observed speed equals V to relative error $\sim 10^{-16}$; (iii) the invariant $|a| f^{3/2} = R\omega^2 \cos i$ holds across all inclinations; and (iv) f and g differ for every $i > 0$. A second script reproduces the same closed forms by independent symbolic computation (SymPy), with every algebraic identity checked to a residual of exactly zero.

This document and the accompanying code form a self-contained worked example in classical mechanics, differential geometry, and symbolic/numerical verification.